

CS4048 — DATA SCIENCE

Lecture 16

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Fall 2025

FAST – NUCES, CFD Campus

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TODAY'S TOPICS

- Hypothesis/significance testing:
 - Critical value: quick recap
 - P-value
 - z-test, t-test, Chi-square test, ANOVA
- A/B testing
- Correlation vs. Covariance



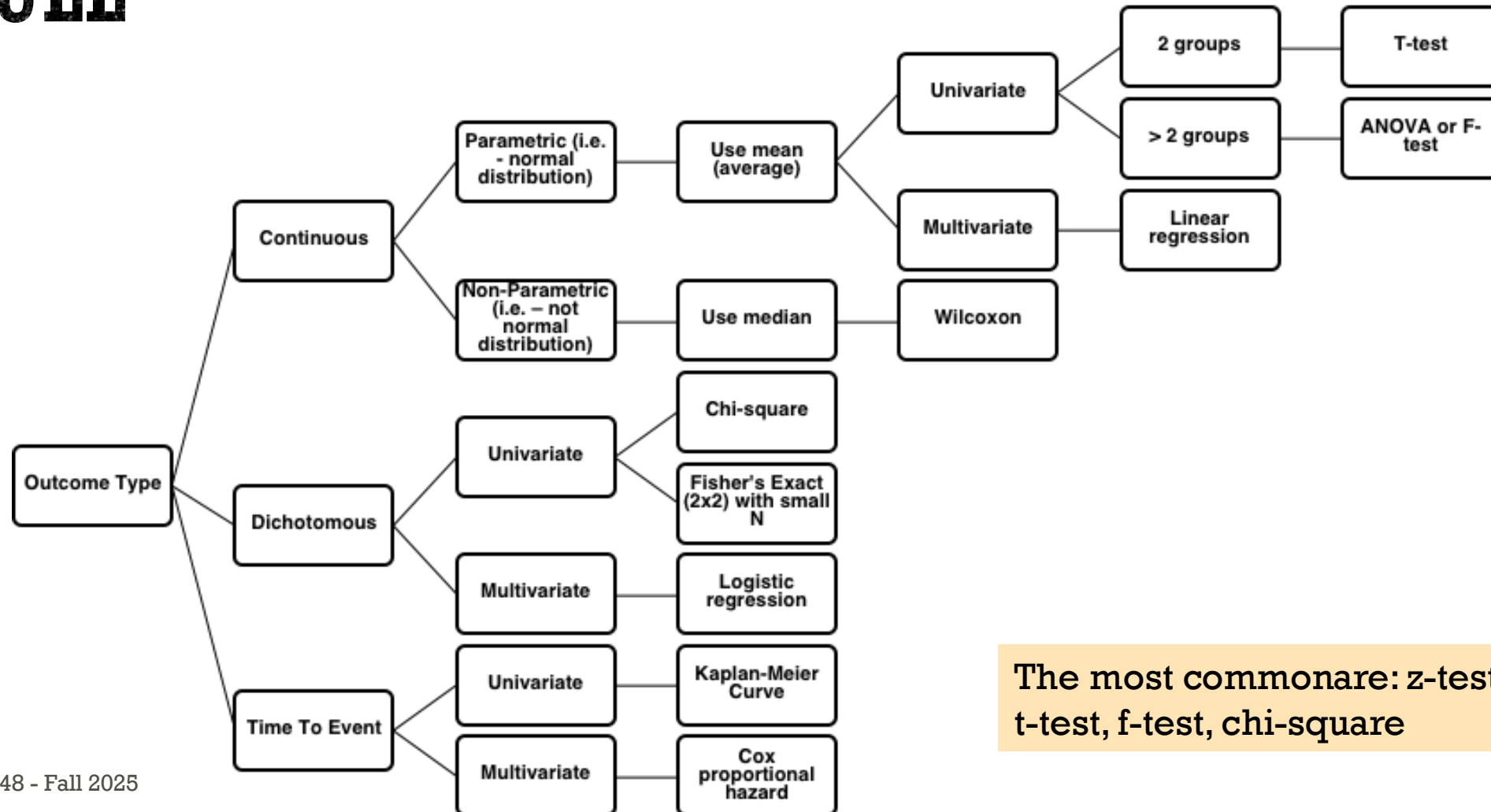
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HYPOTHESIS TESTING STEPS

- Hypothesis tests involve the following steps:
 1. State the null and alternative hypotheses
 2. Select a level of significance
 3. Choose a test criterion (z-test, t-test, chi-square, etc.) & state the decision rule
 4. Calculate the test statistic
 5. Make conclusion (reject or fail to reject the null hypothesis)

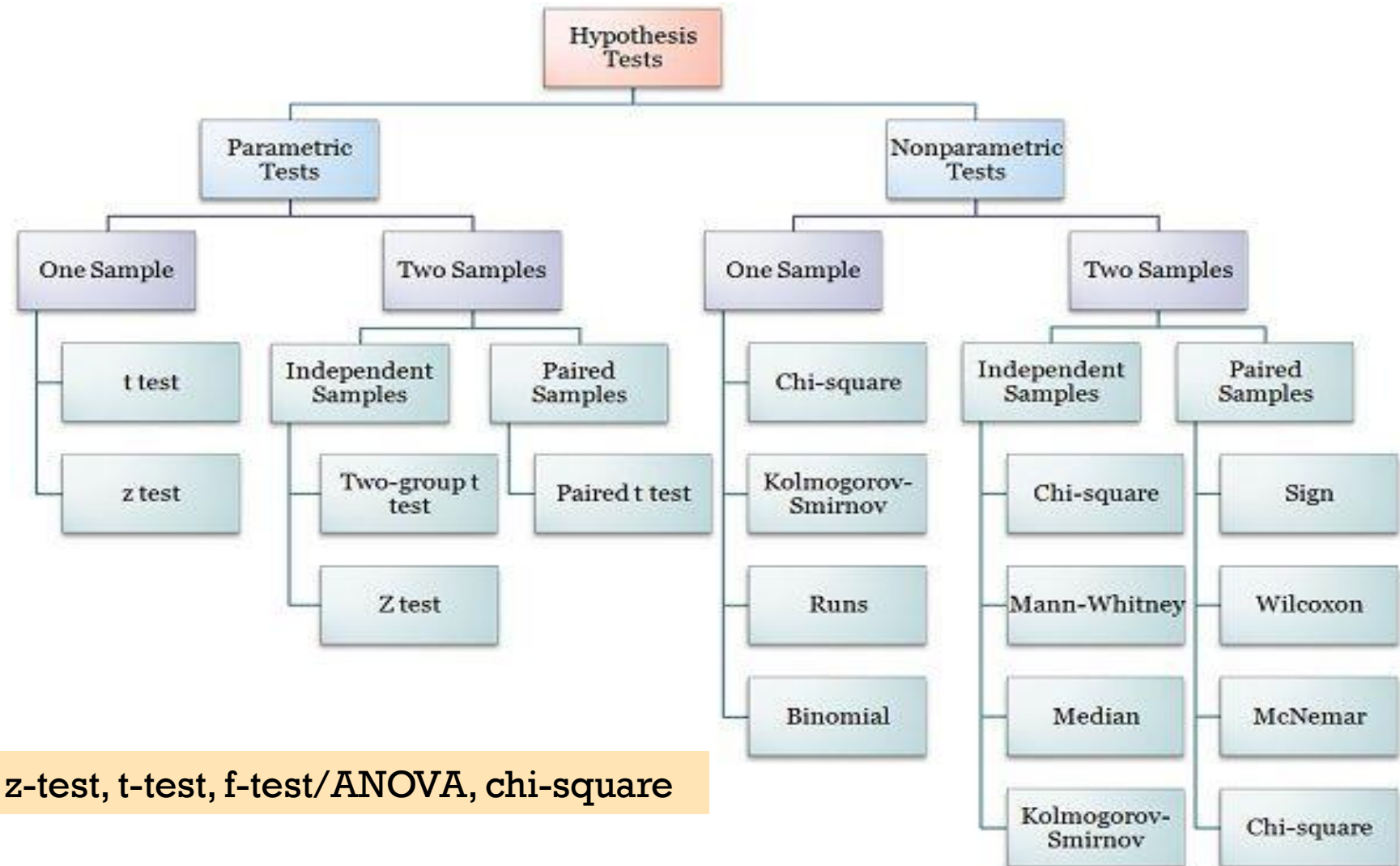
3. CHOOSE TEST CRITERION & DECISION RULE

STATISTICAL TESTS CHEAT SHEET



The most common are: z-test, t-test, f-test, chi-square

3. CHOOSE TEST CRITERION & DECISION RULE



The most commonare: z-test, t-test, f-test/ANOVA, chi-square

3. CHOOSE TEST CRITERION & DECISION RULE

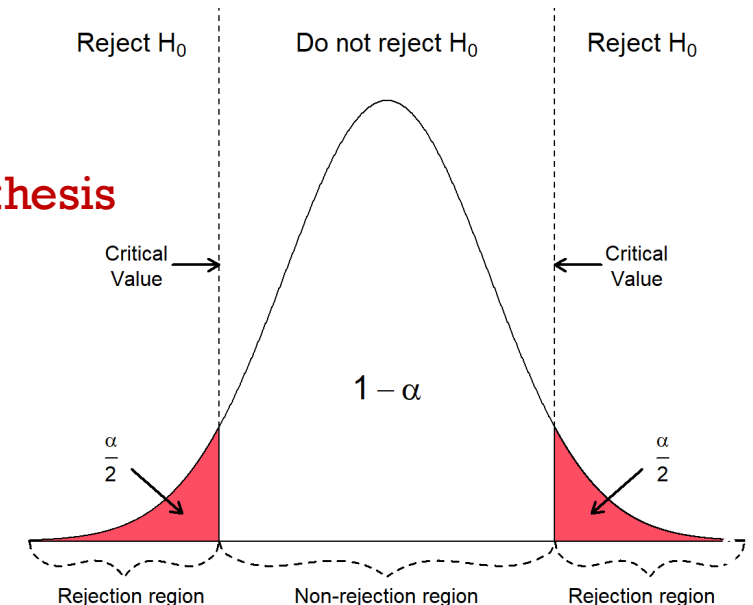
Test statistic	Associated test	Sample size	Information given	Distribution	Test question
z-score	z-test	Two populations or large samples ($n > 30$)	• Standard deviation of the population (this will be given as σ) • Population mean or proportion	Normal	Do these two populations differ?
t-statistic	t-test	Two small samples ($n < 30$)	• Standard deviation of the sample (this will be given as s) • Sample mean	Normal	Do these two samples differ?
f-statistic	ANOVA	Three or more samples	• Group sizes • Group means • Group standard deviations	Normal	Do any of these three or more samples differ from each other?
chi-squared	chi-squared test	Two samples	• Number of observations for each categorical variable	Any	Are these two categorical variables independent?

3. HYPOTHESIS TESTING DECISION

- The decision to reject or not reject the null hypothesis usually is made with reference to the **sampling distribution of a statistic** of some kind (e.g., z-distribution, t-distribution).
- To make a decision whether to reject the null hypothesis, a **test statistic** is calculated.
- There are two approaches how to arrive at that decision:
 1. The critical value approach
 2. The p-value approach

3. HYPOTHESIS TESTING DECISION USING CRITICAL VALUE

- Critical value is a cut-off value that is used to mark the start of a region where the test statistic, obtained in hypothesis testing, is unlikely to fall in.
- The critical value divides the area under the probability distribution curve in **rejection region(s)** and **non-rejection region**.
- **Decision rule:**
 - If the test statistic $>$ critical value $\rightarrow$ **reject** the null hypothesis
 - If the test statistic $\leq$ critical value, **fail to reject** the null hypothesis

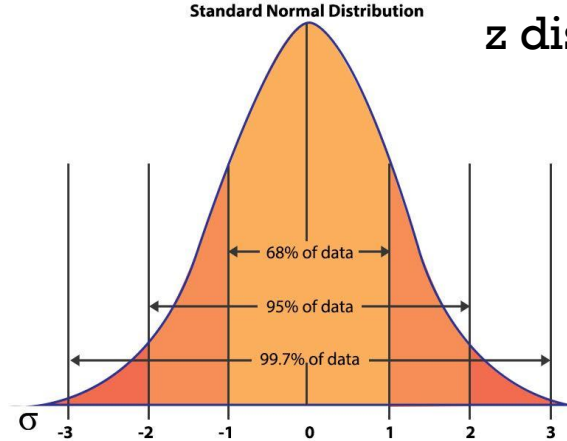
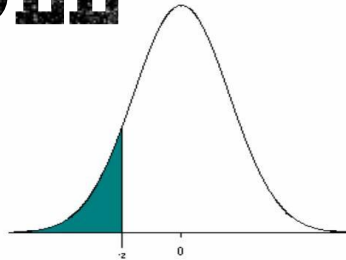


3. HYPOTHESIS TESTING DECISION USING CRITICAL VALUE

- **How to compute critical value?**
 - The critical value is computed based on the given significance level α and the type of probability distribution of the idealized (or null) model.
- We've already done this in confidence interval

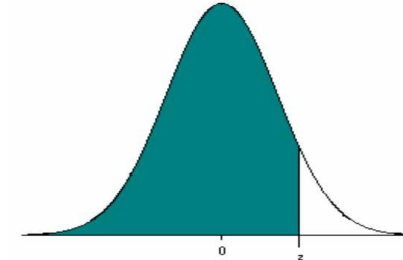
TODO: Find critical value for a two-tailed z-test with $\alpha=0.01$

Z TABLE



z distribution a.k.a “standard normal distribution”

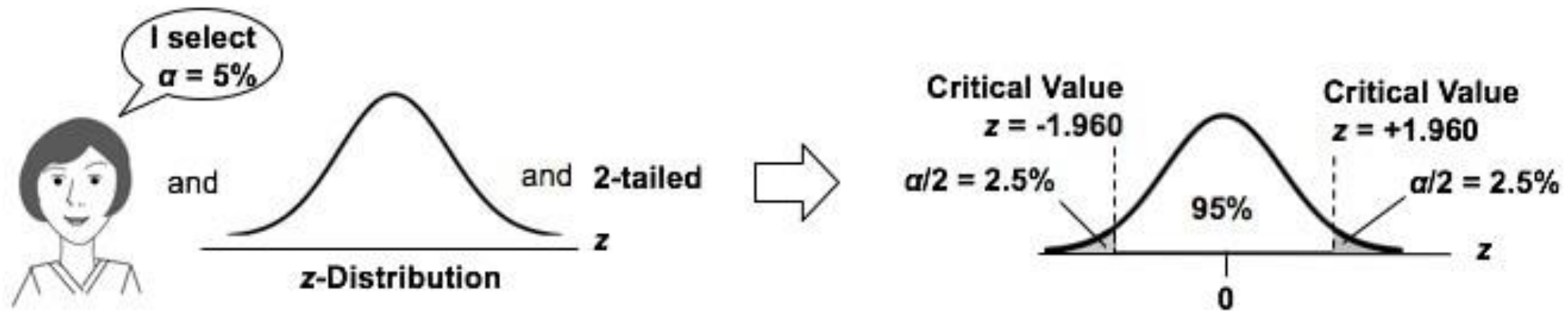
Mean = 0, SD = 1



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

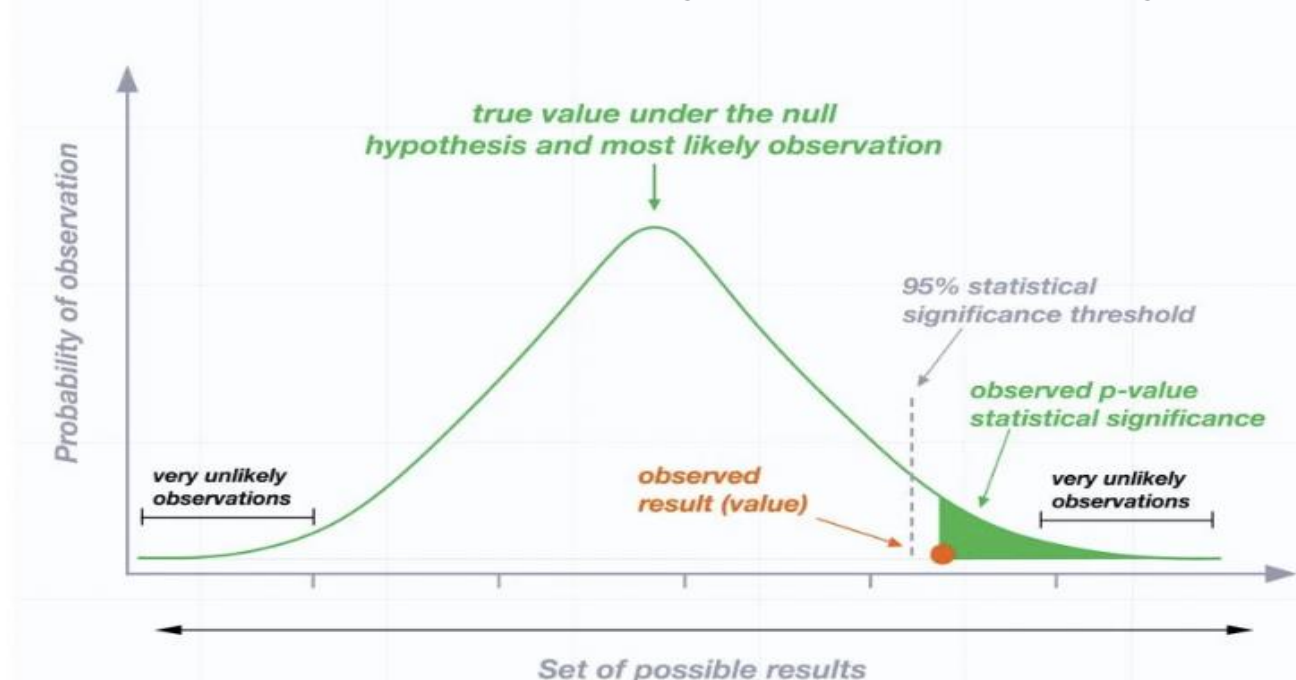
3. HYPOTHESIS TESTING DECISION USING CRITICAL VALUE



To be continued...

3. HYPOTHESIS TESTING DECISION USING P-VALUE

- Another approach to making the decision in hypotheses testing is to compute the p-value associated with the observed value of the test statistic.
- The p-value corresponds to the probability of observing sample data at least as extreme as the actually obtained test statistic.
 - The smaller (closer to 0) the p-value, the stronger is the evidence against the null hypothesis.



3. HYPOTHESIS TESTING DECISION USING P-VALUE

- For the **p-value approach** the **likelihood (p-value)** of the numerical value of the test statistic is **compared to** the **specified significance level (α)** of the hypothesis test.
- **Decision rule:**
 - If the p-value $\leq \alpha \rightarrow$ **reject** the null hypothesis
 - If the p-value $> \alpha \rightarrow$ **fail to reject** the null hypothesis

P-VALUE INTERPRETATION

- P -value answer the question: What is the probability of the observed test statistic ... **when H_0 is true?**
- Thus, smaller and smaller P -values provide stronger and stronger evidence against H_0
- Small P -value $\Rightarrow$ strong evidence



P-VALUE INTERPRETATION

Conventions*

$P > 0.10 \Rightarrow$ non-significant evidence against H_0

$0.05 < P \leq 0.10 \Rightarrow$ marginally significant evidence

$0.01 < P \leq 0.05 \Rightarrow$ significant evidence against H_0

$P \leq 0.01 \Rightarrow$ highly significant evidence against H_0

Examples

$P = .27 \Rightarrow$ non-significant evidence against H_0

$P = .01 \Rightarrow$ highly significant evidence against H_0

* It is *unwise* to draw firm borders for “significance”



3. HYPOTHESIS TESTING DECISION USING P-VALUE

- **How to compute p-value?**
 - Once you compute the test statistic, its corresponding p-value can be found using the distribution table.

3. HYPOTHESIS TESTING DECISION USING P-VALUE

- How to compute p-value?

- Once you compute the test statistic, its corresponding p-value can be found using the distribution table.

- Example: let's assume we are performing a right-tailed test

with sample size >30 , and the computed Z-test statistic is: 0.72.

This means, we need to find the area to the right of z-score of 0.72

which will be our p-value.

$$\text{P-value} = 1 - (\text{area under the z-score}) = 1 - 0.7642 = \mathbf{0.2358}$$

Standard Normal Probabilities

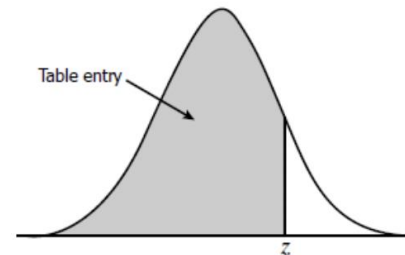


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
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0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177

HYPOTHESIS TESTING STEPS

- Hypothesis tests involve the following steps:
 1. State the null and alternative hypotheses
 2. Select a level of significance
 3. Choose a test criterion (z-test, t-test, chi-square, etc.) & state the decision rule
 4. Calculate the test statistic
 5. Make conclusion (reject or fail to reject the null hypothesis)

4. COMPUTE TEST STATISTIC

- A value computed from a sample data and used to make inference about population data

$$\text{Test Statistic} = \frac{\text{Sample Statistic} - \text{Population}}{\text{Standard Error}} \longrightarrow SE = \frac{\sigma}{\sqrt{(n)}}$$

- For z-test, the (mean) test statistic is:

$$Z_c = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

- For t-test, the (mean) test statistic is:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Test statistic depends on the sample distribution and what you want to compare

<i>Test For</i>	<i>Null Hypothesis (H_0)</i>	<i>Test Statistic</i>	<i>Distribution</i>	<i>Use When</i>
Population mean (μ)	$\mu = \mu_0$	$\frac{(\bar{x} - \mu_0)}{\sigma / \sqrt{n}}$	Z	Normal distribution or $n > 30$; σ known
Population mean (μ)	$\mu = \mu_0$	$\frac{(\bar{x} - \mu_0)}{s / \sqrt{n}}$	t_{n-1}	$n < 30$, and/or σ unknown
Population proportion (p)	$p = p_0$	$\frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$	Z	$n\hat{p}, n(1-\hat{p}) \geq 10$
Difference of two means ($\mu_1 - \mu_2$)	$\mu_1 - \mu_2 = 0$	$\frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	Z	Both normal distributions, or $n_1, n_2 \geq 30$; σ_1, σ_2 known
Difference of two means ($\mu_1 - \mu_2$)	$\mu_1 - \mu_2 = 0$	$\frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	t distribution with $df =$ the smaller of $n_1 - 1$ and $n_2 - 1$	$n_1, n_2 < 30$; and/or σ_1, σ_2 unknown
Mean difference μ_d (paired data)	$\mu_d = 0$	$\frac{(\bar{d} - \mu_d)}{s_d / \sqrt{n}}$	t_{n-1}	$n < 30$ pairs of data and/or σ_d unknown
Difference of two proportions ($p_1 - p_2$)	$p_1 - p_2 = 0$	$\frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	Z	$n\hat{p}, n(1-\hat{p}) \geq 10$ for each group

One sample test

Two samples test

Paired samples test

TERMINOLOGIES

- **One sample test:** compares a single sample's mean to a known or hypothesized population value
 - The test determines if a sample mean is significantly different from a specific value
- **Two samples test:** compares the means of two different, independent groups.
 - The test determines if the means of two separate groups are significantly different from each other.
- **Paired samples test:** each observation in one group is logically connected to another observation in the other group.
 - A paired test determines if there is any significant difference between the two groups with paired observations.

HYPOTHESIS TESTING STEPS

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 4. Calculate the test statistic
 5. Make conclusion (reject or fail to reject the null hypothesis)

5. MAKE CONCLUSIONS

- Remember that we are testing for the NULL hypothesis, that means:
 - We can either reject the NULL hypothesis
 - Or fail to reject the NULL hypothesis
- Never state that we accept the alternative hypothesis.

NOTEBOOK

ANOVA & CHI-SQUARE TESTS

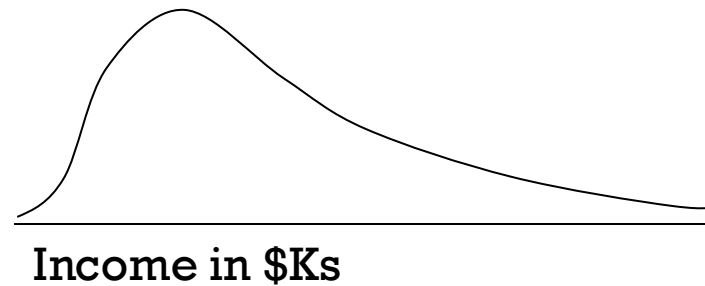
CS4048 - Fall 2025

ANOVA TEST

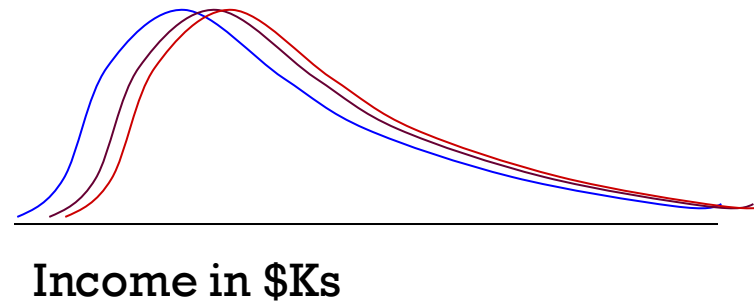
- ANOVA is an inferential statistics technique that allows you to compare the means of two or more groups to determine if there is a significant difference among them.
- NOTE: If you had only two groups to compare, ANOVA would give the same answer as an independent samples t-test.

ANOVA

What if three racial groups had incomes distributed like this in your sample?



All Groups



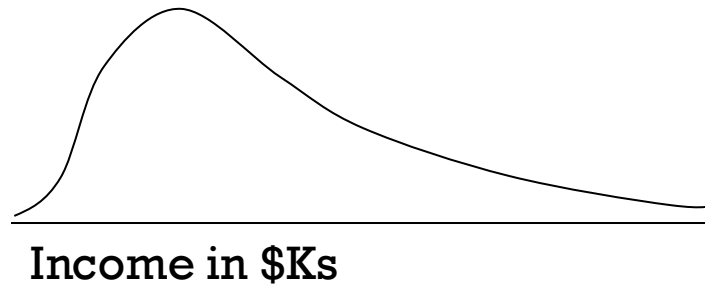
Groups Broken Down

Isn't it conceivable that the differences are due to natural random variability between samples? Would you want to claim they are different in the population?

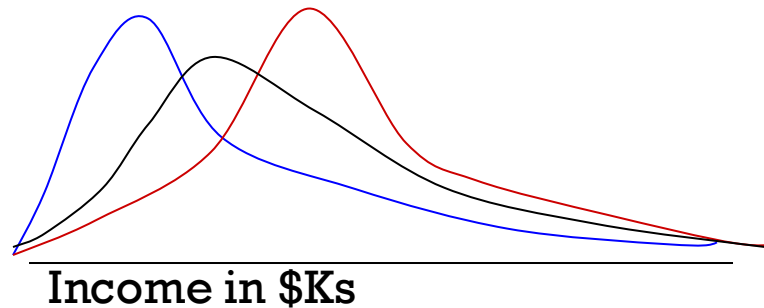


ANOVA

Now...What if three racial groups had incomes distributed like this in your sample?



All Groups Combined



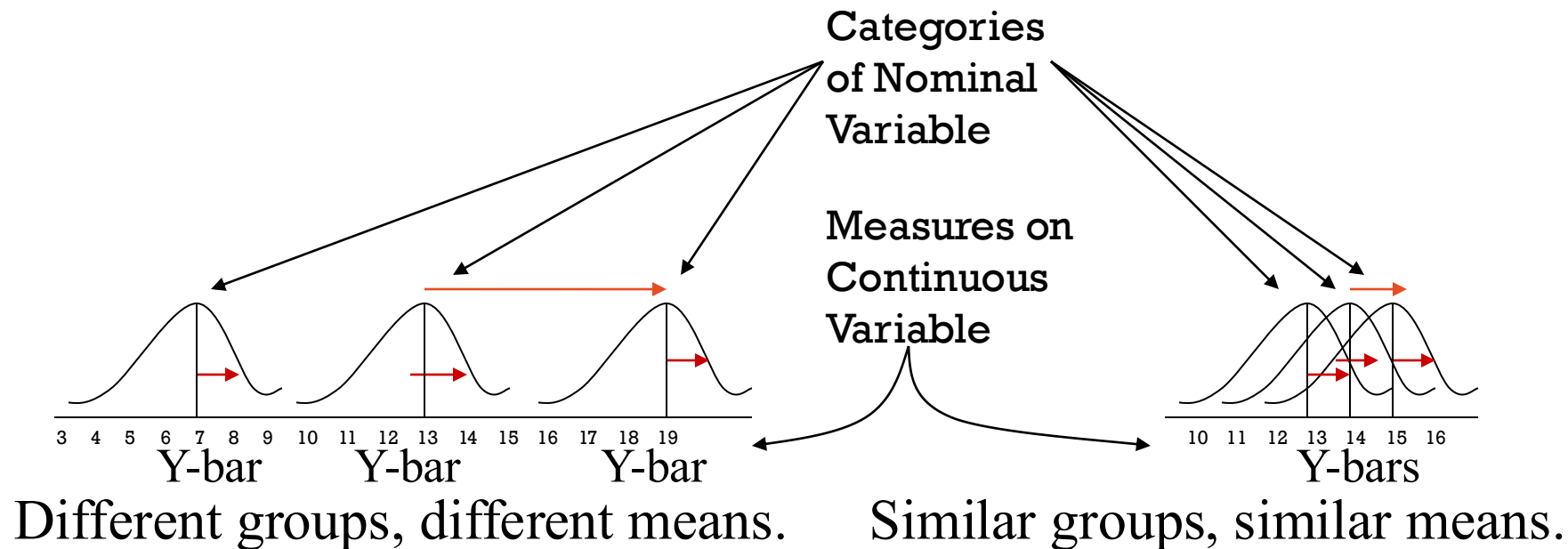
Groups Separated Out

Doesn't it now appear that the groups may be different regardless of sampling variability? Would you feel comfortable claiming the groups are different in the population?



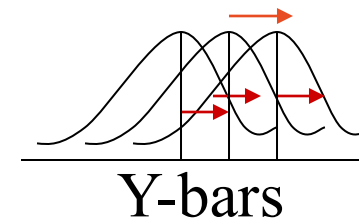
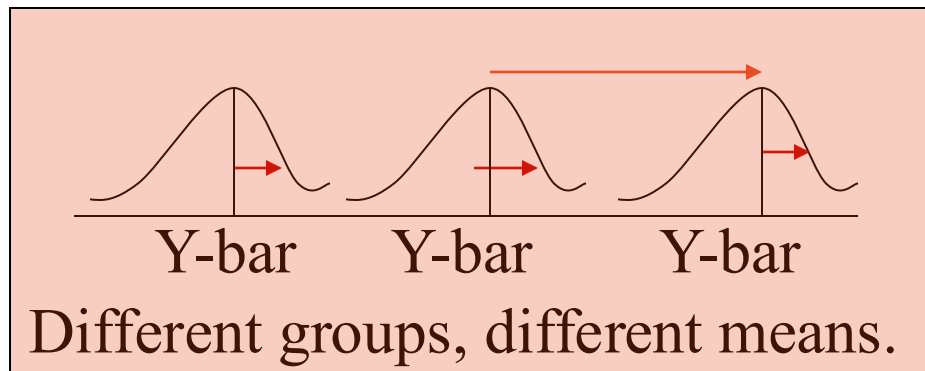
ANOVA

- Conceptually, ANOVA compares the variance **within** groups to the overall variance **between** all the groups to determine whether the groups appear distinct from each other or if they look quite the same.



ANOVA

- When the groups have little variation within themselves, but large variation between them, it would appear that they are distinct and that their means are different.

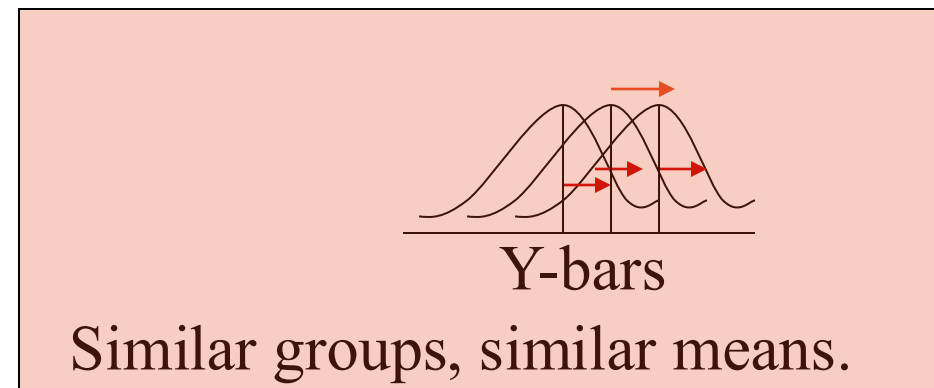
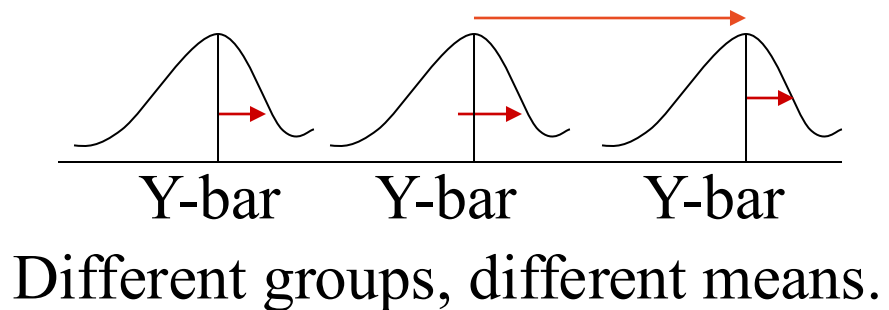


Similar groups, similar means.



ANOVA

- When the groups have a lot of variation within themselves, but little variation between them, it would appear that they are similar and that their means are not really different (perhaps they differ only because of peculiarities of the particular sample).



ANOVA

- Let's call the between groups variation:

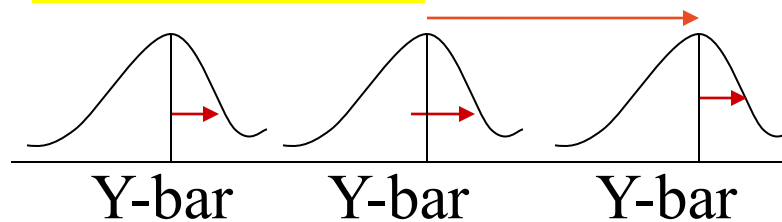
Between Variance: Between Sum of Squares, BSS/df

- Let's call the within groups variation:

Within Variance: Within Sum of Squares, WSS/df

- ANOVA compares Between Variance to Within Variance through a ratio we will call F.

$$F = BSS/WSS$$

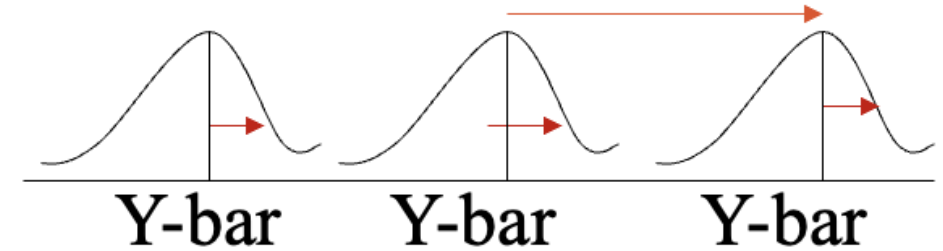


$$F = \frac{BSS/g-1}{WSS/n-g}$$



ANOVA

$$F = BSS/WSS$$



- So what are these BSS and WSS things?
- Remember our friend “Variance?”
- What we are doing is separating out our dependent variable’s overall variance for all groups into that which is attributable to the **deviations of groups’ means from the overall mean (B)** and **deviations of individuals’ scores from their own group’s mean (W)**.



ANOVA

Variance:

Deviation

Squared Deviation

Sum of Squares

Variance

$$Y_i - \bar{Y}$$

$$(Y_i - \bar{Y})^2$$

$$\Sigma(Y_i - \bar{Y})^2$$

$$\Sigma(Y_i - \bar{Y})^2$$

$$n - 1$$

(Standard Deviation)

$$\sqrt{\frac{\Sigma(Y_i - \bar{Y})^2}{n - 1}}$$

ANOVA SEPARATING VARIANCE

Take your continuous variable and separate scores into groups according to your nominal variable

Between Variance

$$\begin{aligned}\text{Deviation :} & Y\text{-bar}_g - Y\text{-bar}_{\text{big}} \\ \text{Squared Deviation:} & (Y\text{-bar}_g - Y\text{-bar}_{\text{big}})^2 \\ \text{Weight by number} & \\ \text{of people in each} & \\ \text{group:} & n_g * (Y\text{-bar}_g - Y\text{-bar}_{\text{big}})^2 \\ \\ \text{Sum of Squares:} & \\ & \Sigma (n_g * (Y\text{-bar}_g - Y\text{-bar}_{\text{big}})^2) \\ \text{Variance:} & \frac{\Sigma (n_g * (Y\text{-bar}_g - Y\text{-bar}_{\text{big}})^2)}{g - 1} \\ & \dots \text{ or BSS}/g-1\end{aligned}$$

$Y\text{-bar}_g$ = Each Group's Mean

g = # of Groups

Y_i = Each Data Point in a Group

Within Variance

$$\begin{aligned}\text{Deviation :} & Y_i - Y\text{-bar}_g \\ \text{Squared Deviation:} & (Y_i - Y\text{-bar}_g)^2 \\ \\ \text{Sum of Squares:} & \Sigma (\Sigma (Y_i - Y\text{-bar}_g)^2) \\ \text{Variance:} & \frac{\Sigma (\Sigma (Y_i - Y\text{-bar}_g)^2)}{n - g} \\ & \dots \text{ or WSS}/n-g\end{aligned}$$

$Y\text{-bar}_{\text{big}}$ = Overall Mean

n_g = # of Cases in Each Group

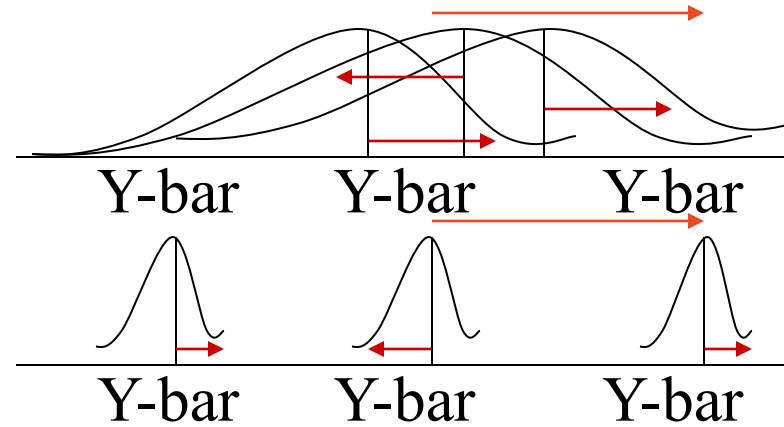
n = # of Cases in Overall Sample

ANOVA

$$F = \frac{\text{BSS}/g-1}{\text{WSS}/n-g}$$

As WSS gets larger, F gets smaller.

As WSS gets smaller, F gets larger.

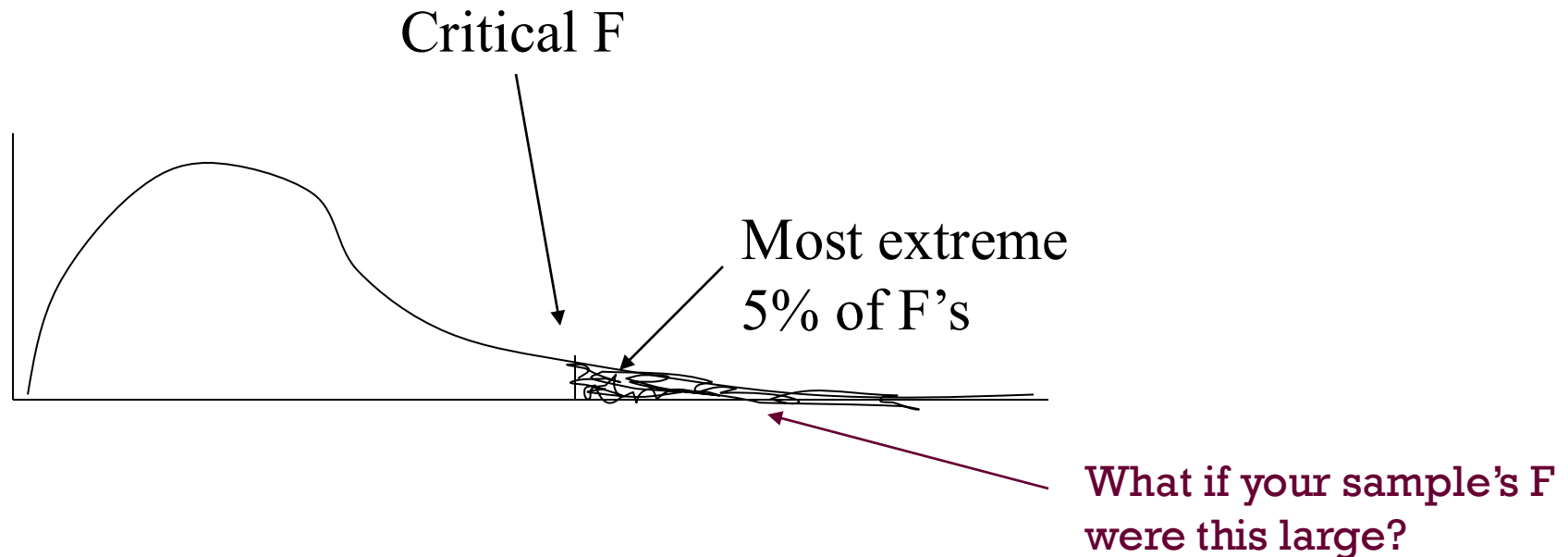


So, as F gets smaller, the groups are less distinct. As F gets larger, the groups are more distinct.

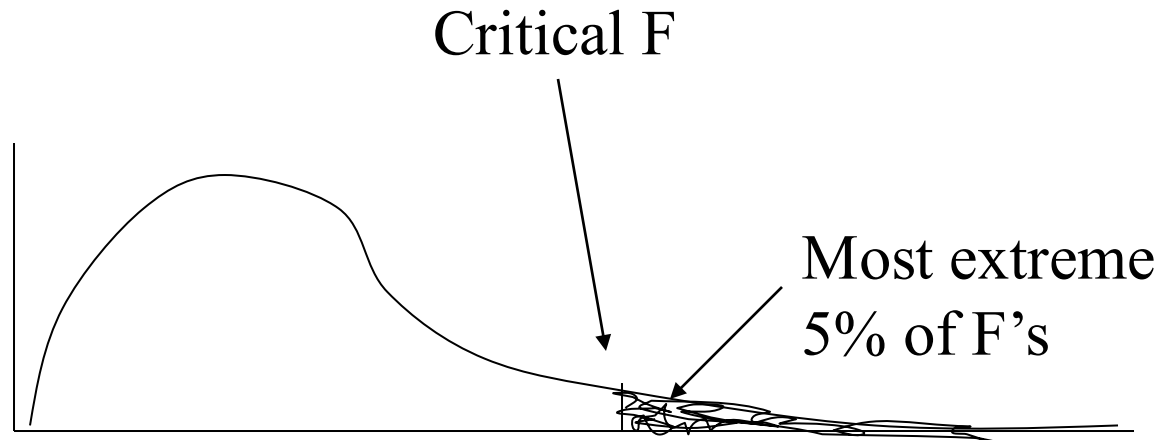


ANOVA

- The F distribution is like the normal curve, t distribution, and chi-squared distribution: we know that there is a critical F that demarcates the value beyond which the values of the rarest 5% of F's will fall.



ANOVA



- One other thing to note is that, like chi-squared, there are different F distributions depending on the degrees of freedom (df) of the BSS and the WSS. E.g., $F(g-1, n-g)$.
- BSS df: # of groups in your nominal variable minus 1
- WSS df: # of cases in sample minus number of groups



ANOVA

Null: Group means are equal; $\mu_1 = \mu_2 = \dots = \mu_g$

Alternative: At least one group's mean is different

So... if F is larger than the critical F:

Reject the Null in favor of the alternative!



CHI-SQUARE TEST

- A **non-parametric** test that is used to measure the association between two categorical variables.
 - Since we only require that the two variables are categorical, there are no assumptions required on the variables in the Chi-Square test.
- We use it when we have observed frequencies of a categorical variable (e.g., male vs female, healthy vs sick).

CHI-SQUARE TEST

There are 2 types of chi-square tests:

1. The **chi-square goodness of fit test** is used to compare the observed frequencies in a data sample with the frequencies based on some prior expectation – either empirical or theoretical.
2. The **chi-square test of independence** assesses whether observed frequencies are dependent on (i.e., contingent on) certain conditions.

CHI-SQUARE TEST

- A frequency distribution table / contingency table shows how observations are distributed between different groups (i.e., the number of observations in each group).

Example: Bird species at a bird feeder

Frequency of visits by bird species at a bird feeder during a 24-hour period

Bird species	Frequency
House sparrow	15
House finch	12
Black-capped chickadee	9
Common grackle	8
European starling	8
Mourning dove	6

A **chi-square goodness of fit test** can test whether these observed frequencies are significantly different from what was expected, such as equal frequencies.

Null hypothesis (H_0): The bird species visit the feeder in the **same** proportions as the average over the past five years.

Alternative hypothesis (H_1): The bird species visit the feeder in **different** proportions from the average over the past five years.

CHI-SQUARE TEST

- A frequency distribution table / contingency table shows how observations are distributed between different groups (i.e., the number of observations in each group).

Example: Handedness and nationality

Contingency table of the handedness of a sample of Americans and Canadians

	Right-handed	Left-handed
American	236	19
Canadian	157	16

A **chi-square test of independence** can test whether these observed frequencies are significantly different from the frequencies expected if handedness is unrelated to nationality.

Null hypothesis (H_0): The proportion of people who are left-handed is **the same** for Americans and Canadians.

Alternative hypothesis (H_1): The proportion of people who are left-handed **differs** between nationalities.

CHI-SQUARE (χ^2) STATISTIC

- Both of Pearson's chi-square tests use the same formula to calculate the chi-square (χ^2) statistic:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where:

χ^2 is the chi-square statistic

Σ is the summation operator (i.e., it “takes the sum of”)

O is the observed frequency

E is the expected frequency

The larger the difference between the observations and expectations ($O - E$ in the equation), the bigger the chi-square statistic will be.

The chi-square statistic gets compared with a critical value using a chi-square critical value table or statistical software.

READING MATERIAL

- Chapter#3: Principle of Data Science
- Chapter#1, 2: Practical Statistics [Bruce]
- An article: <https://www.kdnuggets.com/2021/03/statistics-data-scientists-should-know.html>
- Z-test in ML: <https://mlvidhya.com/blog/hypothesis-testing/z-test-in-hypothesis-testing>
- A useful resoure on inferential statistics:
<https://www.statisticsfromatoz.com/blog/category/new-video/4>